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Paleomagnetism of the 1.88-Ga Sokoman Formation in the Schefferville–Knob Lake area, Québec, Canada, and implications for the genesis of iron oxide deposits in the central New Québec Orogen

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Abstract

Thirty-nine oriented block samples of iron-formation were collected at 13 sites, including opposite limbs of major folds, from the 1.88-Ga Sokoman Formation (Knob Lake Group) in the Schefferville–Knob Lake area of the central New Québec Orogen, northern Québec. The samples assayed up to 80.24% Fe_2O_3T (54.08% Fe), implying Fe-enrichment of the iron-formation up to ore grade. Anisotropy of magnetic susceptibility measurements on 245 standard specimens indicate a well preserved bedding-parallel fabric in the iron-formation, suggesting minimal alteration of the magnetic mineralogy since deposition and/or a mimetic secondary magnetic mineralogy. The iron-formation has not been internally deformed since the magnetic mineralogy was established. Analyses by variable-field translation balance and X-ray diffraction showed that the predominant magnetic mineral is hematite but a small amount of magnetite also is present in most samples. Following low-temperature pre-treatment as appropriate, stepwise thermal and alternating-field demagnetization of 218 specimens revealed a low-temperature, post-folding component (maximum $T_{ub} \approx 400\,^{\circ}C$, $D = 27.1^{\circ}$, $I = 20.1^{\circ}$, $\alpha_{95} = 10.9^{\circ}$, from seven sites; pole position of 40.6°S, 257.0°E), and components carried by magnetite (maximum $T_{ub} \approx 580\,^{\circ}C$, $D = 35.8^{\circ}$, $I = 3.9^{\circ}$, $\alpha_{95} = 9.1^{\circ}$, from 10 sites; pole position of 29.6°S, 250.9°E) and hematite (maximum $T_{ub} \approx 680\,^{\circ}C$, $D = 40.0^{\circ}$, $I = 1.6^{\circ}$, $\alpha_{95} = 18.6^{\circ}$, from seven sites; pole position of 26.8°S, 247.0°E). The components carried by magnetite and hematite are pre-, syn- and post-folding depending on the sampling site, indicating that the magnetization was acquired continuously with deformation in the New Québec Orogen at 1.84–1.83 Ga. No evidence was found for acquisition of magnetization during the Mesozoic, when many of the iron oxide orebodies in the Schefferville–Knob Lake area are thought to have formed. Our findings imply that an episode of Fe-enrichment of iron-formation in the Sokoman Formation involved the circulation of hydrothermal fluids related to late Paleoproterozoic orogenesis. Such orogenic circulation of fluids may have contributed to the development of hematitic orebodies in the central New Québec Orogen.
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1. Introduction

Most studies of the paleomagnetism of Paleoproterozoic iron-formations have focused on the identifi-

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cation of overprints related to the development of secondary iron oxides and implications for the timing and genesis of related iron oxide orebodies. In a pioneering study, Gross and Strangway (1961) determined natural remanent magnetization (NRM) directions for two iron oxide orebodies that occur in Precambrian banded iron-formation in Mauritania, concluding that one orebody is Precambrian and the other formed by surface leaching in more recent times. Other paleomagnetic investigations of hematite orebodies within Archean and Proterozoic banded iron-formations in Western Australia and South Australia (Porath, 1967; Chamalaun and Porath, 1968; Porath and Chamalaun, 1968) suggested a Precambrian age for the orebodies. This finding was supported by later studies integrating available geological and paleomagnetic evidence, which implied that the giant Hamersley microplaty hematite orebodies in Western Australia formed during or soon after Paleoproterozoic deformation (Clark and Schmidt, 1993; Li et al., 1993; Schmidt and Clark, 1994; Powell et al., 1999). In North America, paleomagnetic studies suggested Proterozoic ages for hard hematite and goethite orebodies in Minnesota, Michigan and Ontario (Symons, 1967a,b).

The present work continues our investigation of the rock magnetism and paleomagnetism of Paleoproterozoic iron-formations that aims to elucidate their paleogeography and metamorphic history (Williams et al., 2003; Schmidt and Williams, 2003). Here we study iron-formation from the 1.88-Ga Sokoman Formation in northern Québec and discuss the implications of our results for the genesis of iron oxide deposits in the central New Québec Orogen.

2. Geological setting

2.1. Knob Lake Group, New Québec Orogen

The New Québec Orogen (formerly called the Labrador Trough; Hoffman, 1988) of northeastern Québec and Labrador is a ~1000 km long Paleoproterozoic fold and thrust belt marking the collision between the Archean Superior Province to the southwest and the largely Archean Rae Province to the northeast (Fig. 1). The western part of the central New Québec Orogen ('Schefferville Zone') preserves the most complete and least deformed portion of the autochthonous succession (Wardle et al., 1990). In the Schefferville Zone, Archean gneisses of the Superior Province are unconformably overlain by the Knob Lake Group, which consists of two sedimentary cycles (Fig. 2). Cycle 1 is a marine-shelf succession comprising the Le Fer siltstone, Denault dolostone, Dolly siltstone–shale and Fleming chert breccia. Cycle 2 begins with a transgressive quartzarenite (Wishart Formation), followed by the Ruth shale and iron-formation of the Sokoman Formation deposited on a subsiding marine ramp and conformably overlain by the flyschoid Menihek Formation.

The Sokoman Formation and correlative iron-formation that occur throughout the length of the New Québec Orogen constitute one of the most extensive iron-formations in the world. In the Schefferville–Knob Lake area (centered on 54°48′N, 66°49′W) the Sokoman Formation is between 120 and 240 m thick and a typical cherty, Superior-type iron-formation, containing peloids and ooliths and commonly displaying thin and irregular bedding as well as cross-bedding and intraformational conglomerates (Stubbins et al., 1961; Zajac, 1974). The formation is divisible stratigraphically into three regionally extensive members: a lower member (10–30 m) of silicate–carbonate (siderite) and shale, a middle member (90–150 m) of mainly cherty hematite iron-formation containing 30–70% iron oxides, and an upper member (25–60 m) with silicate–carbonate (Stubbins et al., 1961; Chauvel and Dimroth, 1974).

Igneous activity in the Schefferville Zone commenced during deposition of the lower Knob Lake Group, reached a climax during deposition of the Sokoman Formation and ended when Menihek deposition commenced. Mafic to felsic volcanic rocks of the Nimish Formation occur at the base and top of the Sokoman Formation east of Schefferville, and episodic Nimish magmatism occurred throughout the entire interval of iron-formation deposition (Zajac, 1974; Findlay et al., 1995). The close spatial and temporal relationship between Sokoman deposition and local volcanic activity suggests that the iron in the Sokoman iron-formation is volcanogenic (Seguin, 1976b). Cameron (1983) concluded that sulfur isotope data for the Ruth shale and Menihek Formation are consistent with a hydrothermal exhalative origin for the iron and silica in iron-formation of the associated Sokoman Formation.

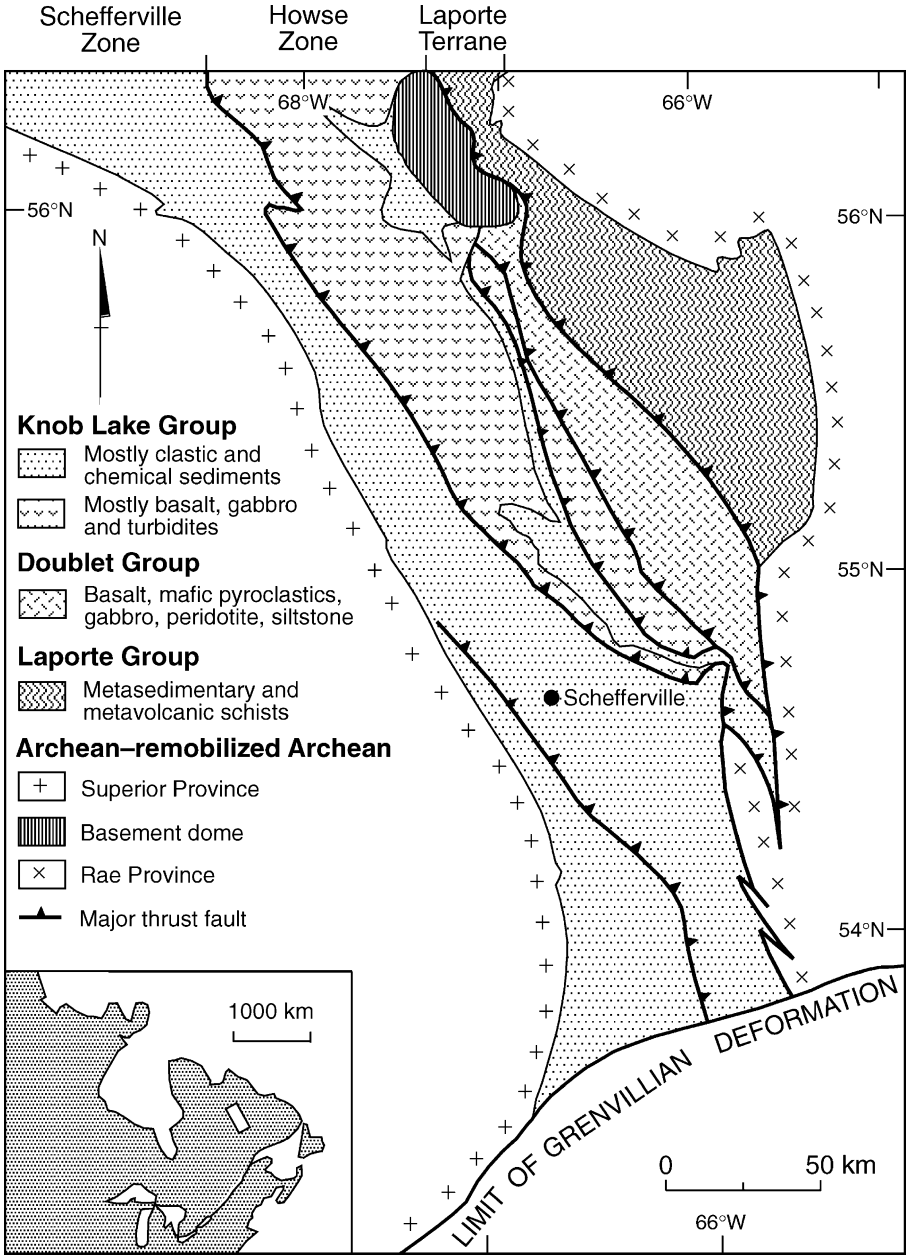


Fig. 1. Lithotectonic subdivision of the central New Québec Orogen (modified after Findlay et al., 1995).

A U–Pb zircon age of 1880 ± 2 Ma for a carbonatite dyke coeval with the Sokoman Formation provides the best estimate for the timing of Sokoman deposition (Findlay et al., 1995). A U–Pb zircon age of 1878 ± 1 Ma for syenite cobbles from the Nimish igneous suite (Findlay et al., 1995) is equivalent, within error, to the above estimate for the age of the Sokoman Formation. A gabbro sill within the overlying

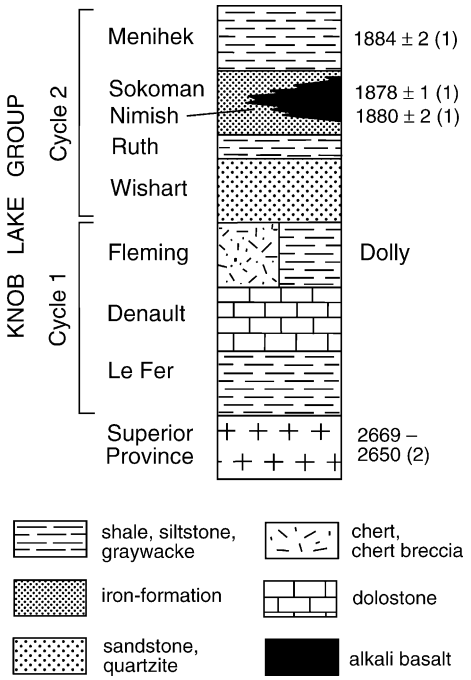


Fig. 2. Generalized stratigraphy of the Knob Lake Group, New Québec Orogen (modified after Findlay et al., 1995). Ages (Ma) from (1) Findlay et al. (1995), (2) Mortensen and Percival (1987).

Menihek Formation ~50 km north–northeast of Schefferville gave a U–Pb zircon age of 1884 ± 2 Ma, suggesting to Findlay et al. (1995) that the Sokoman and Menihek formations are diachronous, younging to the west.

Thrusting and metamorphism in the New Québec Orogen in response to the Rae-Superior convergence and collision occurred at 1840–1829 Ma (Machado, 1990). In the west, low- to medium-grade supracrustal rocks were transported westward over the Archean Superior basement in a foreland fold and thrust belt marked by a series of imbricate thrusts (Wardle et al., 1990). Metamorphic grade in the New Québec Orogen increases eastward from subgreenschist facies in the Schefferville Zone to upper amphibolite–granulite facies in the eastern part of the orogen (Dimroth and Dressler, 1978; Hoffman, 1988). The Knob Lake Group in the Schefferville–Knob Lake area is within the subgreenschist zone, implying temperatures of <300 °C (Zajac, 1974).

2.2. Iron oxide orebodies, Schefferville–Knob Lake area

At least 45 hematite–goethite orebodies occur within the 20 km wide Knob Lake Range that runs for 100 km northwest through the Schefferville–Knob Lake area (Stubbins et al., 1961; Gross, 1968). From 1954 to 1982 the Iron Ore Company of Canada produced ~250 million tonnes of direct shipping iron ore (>50% Fe) and treat rock from seven open-pit mines within 8 km of Schefferville, and at its highest rate shipped 10 million tonnes per year of direct shipping ore to Sept Iles on the Gulf St. Lawrence (Engineering and Mining Journal, 1982; Neal, 2000). About 200 million tonnes of iron resources remain in the 45 deposits.

Three principal types of bedded ore formed in situ from Sokoman iron-formation and Ruth shale. Blue ore, consisting mostly of dark gray-black hematite with lesser amounts of red hematite, martite and brown goethite, is derived mainly from the cherty iron oxide member of the Sokoman Formation and forms the richest (mean 54.5% Fe) and most abundant (66%) deposits. Yellow to brown ore (mean 50.0% Fe and 20% of deposits) comprises mainly goethite and hydrous iron oxides and dark brown martite and is derived from the silicate–carbonate member of the Sokoman Formation. Red ore (mean 48.0% Fe, 14% of deposits) consists mainly of red hematite, goethite, soft aluminous silicates, and chert and is derived mainly from the Ruth shale. Rubble ores >60 m thick that are preserved in minor, down-faulted basins were derived from some of the bedded residual orebodies and contain early Late Cretaceous plant remains (Dorf, 1959).

The bedded ores are believed to have formed through the leaching of silica and carbonates, the formation of martite and goethite, and the redistribution of iron oxides by circulating meteoric waters during prolonged and deep chemical weathering in Mesozoic times (Stubbins et al., 1961; Gross, 1968). The presence of Cretaceous rubble ores suggests, however, that at least some of the bedded residual orebodies may be substantially older. Stubbins et al. (1961) noted that hydrothermal alteration may better account for some of the observed mineralogical and chemical changes, but concluded that the absence of intrusions in close proximity to the orebodies made hydrothermal activity appear unlikely.

3. Previous paleomagnetic studies of the Knob Lake Group

Previous paleomagnetic data for the Knob Lake Group (Table 1) are sparse and restricted to early investigations of the Sokoman Formation and igneous rocks from the Nimish and Menihek formations.

Seguin (1976a) carried out a paleomagnetic study of the Sokoman Formation and diabase dykes cutting the iron-formation in the Schefferville–Knob Lake area. Following alternating field (AF) and thermal demagnetization, he claimed to have identified a pre-folding NRM direction for the iron-formation with a paleopole position of 33°S, 256°E, and a post-folding NRM direction for combined data for diabase and iron-formation with a paleopole position of 16°N, 258°E.

Seguin (1979) studied the paleomagnetism of metamorphosed iron-formation from the southern New Québec Orogen that is adjacent to or within the Grenville Province. Following AF demagnetization, he obtained a direction interpreted as an overprint related to the Grenville Orogeny with a paleopole position of 46°S, 186°E, and a direction interpreted as primary with a paleopole position of 29°N, 263°E.

Park (1977) conducted a reconnaissance paleomagnetic investigation of the central New Québec Orogen 120–220 km north of the Schefferville–Knob Lake area, sampling the Sokoman Formation at four sites. Directions for metasedimentary iron-formation at about the Curie point of magnetite were widely scattered (D = 03° to 219°, I = 07° to 65°), with some persisting to the Curie point of hematite, and Park (1977) regarded the thermal demagnetization curves as indicating a thermal remanent magnetization (pTRM) acquired by mild heating during deep burial. Other data for the Knob Lake Group provided by Park (1977) comprised three directions for the Menihek Formation: an A direction (D = 154°, I = 21°, N = 1 site; pole position of 19°S, 318°E) with an unblocking temperature of ∼600 °C and thought to have been acquired soon after folding; a B direction (D = 195°, I = 57°, N = 5, α_{95} = 16°; pole position of 05°N, 279°E) with an unblocking temperature of 300–500 °C and possibly acquired during uplift following folding; and a much younger C direction (D = 298°, I = 72°, N = 4, α_{95} = 16°; pole position of 57°N, 228°E) with an unblocking temperature of <300 °C.

Seguin (1976b) sampled greenstones and ironstone-bearing agglomerates from the Nimish Formation at 28 sites north and northeast of Schefferville. Stepwise thermal demagnetization to 550 °C and unfolding (bedding dip-correction) yielded a pole position of 05°S, 268°E, which he interpreted as an initial thermoremanent component. Stepwise AF demagnetization to 700 Oe and unfolding yielded a pole position of 31°S, 239°E, that Seguin (1976b) related to the ∼1750-Ma 'Hudsonian' orogeny.

Seguin (1976c) studied gabbros intrusive into the Menihek Formation and metabasalts intercalated with that formation in the central New Québec Orogen 150 km north of Schefferville. After AF demagnetization to 1100 Oe he identified a combined direction that he regarded as pre-folding and which yielded a paleopole position of 02°N, 205°E.

These early investigations of the Knob Lake Group including the Sokoman Formation have yielded some paleomagnetic data of dubious reliability and conflicting results (see also Section 7.1), indicating the necessity for further study of the Sokoman Formation.

4. Sampling of the Sokoman Formation

4.1. Sampling sites

Three oriented ∼2 kg block samples of iron-formation were collected from each of 13 sites in the middle member of the Sokoman Formation between Paulette Lake 11 km south of Schefferville and Lake Attikamagen 15 km northeast of Schefferville (Fig. 3). Opposite limbs of major folds were sampled to provide fold tests. Orientation was carried out using a tool that accommodates both a sun compass and a magnetic compass. The mean of the local declination thus measured agreed within error with the published value of 25°W.

Site details and mean attitude of bedding are as follows (rock color designations from the Geological Society of America Rock-Color Chart, 1963):

- Site 01. By road 1 km south of Paulette Lake, at 54°42.0′N, 66°45.2′W. Grayish brown (5 YR

Table 1
Previous paleomagnetic data for the Knob Lake Group, New Québec Orogen

Rock unit	Locality	D (°)	I (°)	Pole position	Interpreted component	Reference
Menihek Formation	150 km north of Knob Lake	268	06	02°N, 205°E	Pre-folding, 1100 Oe	Seguin (1976c)
Menihek Formation	120–220 km north of Knob Lake	154	21	19°S, 318°E	A, ∼600 °C	Park (1977)
Menihek Formation	120–220 km north of Knob Lake	195	57	05°N, 279°E	B, 300–500 °C	Park (1977)
Menihek Formation	120–220 km north of Knob Lake	298	72	57°N, 228°E	C, <300 °C	Park (1977)
Sokoman Formation	Knob Lake area	–	–	33°S, 256°E	Pre-folding, NRM	Seguin (1976a)
Sokoman Formation	Knob Lake area	–	–	16°N, 258°E	Post-folding, NRM	Seguin (1976a)
Sokoman equivalent (?)	Grenville Province	–	–	46°S, 186°E	Grenvillian overprint, 100–400 Oe	Seguin (1979)
Sokoman equivalent (?)	Grenville Province	54	−78	29°N, 263°E	Primary, 400–700 Oe	Seguin (1979)
Sokoman Formation	120–220 km north of Knob Lake	03–219	07 to 65	–	pTRM, ∼580–680 °C	Park (1977)
Nimish Formation	North and northeast of Schefferville	209	52	05°S, 268°E	Initial TRM, 550 °C	Seguin (1976b)
Nimish Formation	North and northeast of Schefferville	226	−18	31°S, 239°E	‘Hudsonian’ overprint, 700 Oe	Seguin (1976b)

D: declination, I: inclination.

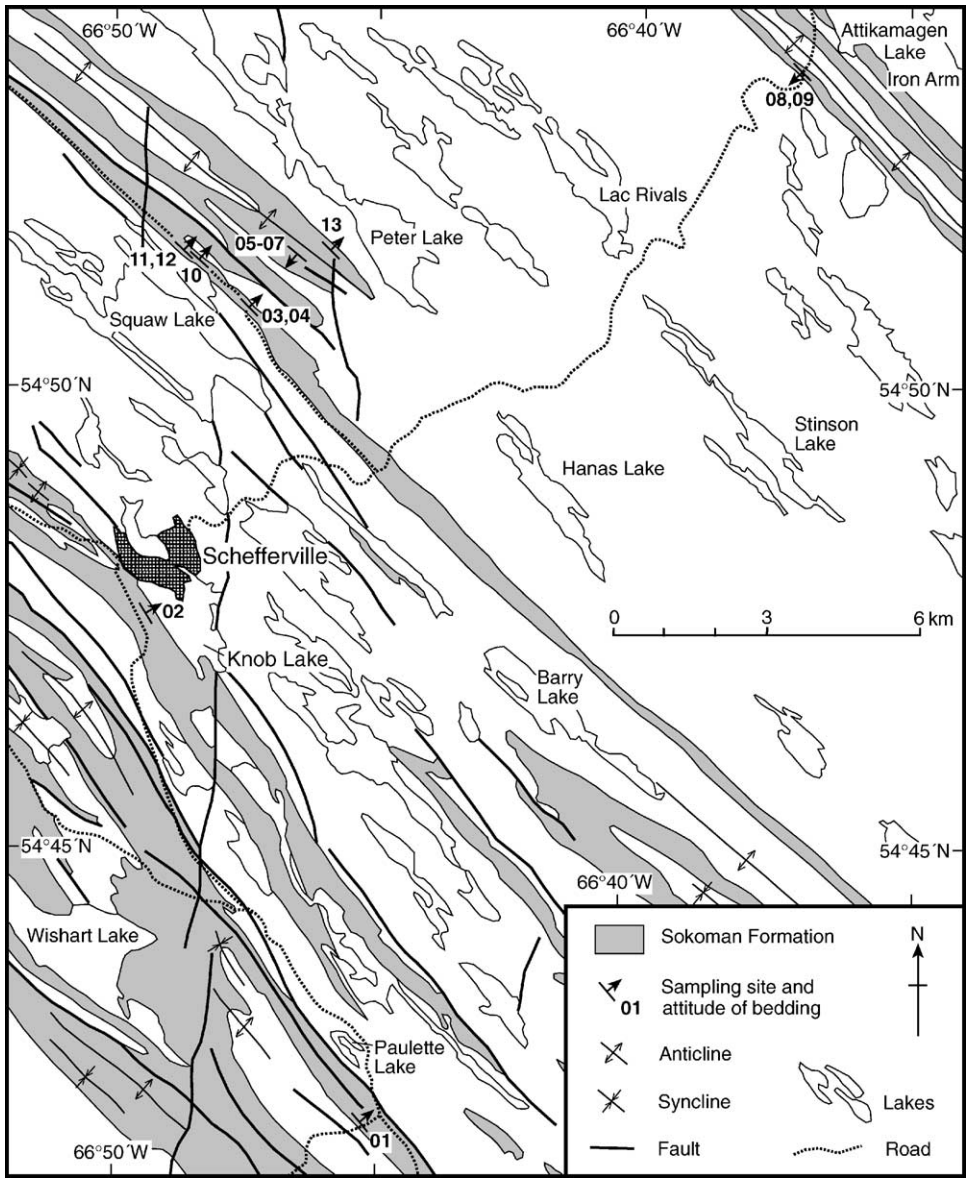


Fig. 3. Geological sketch map showing the distribution and structure of the Sokoman Formation in the Schefferville–Knob Lake area between Paulette Lake and Lake Attikamagen and the location of the 13 paleomagnetic sampling sites. Unshaded areas represent undifferentiated Knob Lake Group. Modified from Zajac (1974, Fig. 86) and Wardle (1982).

3/2) to yellowish brown (10 YR 5/4), peloidal and banded iron-formation. Dip 30° to the northeast.

- Site 02. Ridge on west side of Knob Lake, at 54°47.5′N, 66°49.6′W. Blackish red (5 R 2/2) jaspilitic iron-formation. Dip 40° to the east-northeast.
- Sites 03 and 04. By road on east side of Squaw Lake, at 54°50.8′N, 66°47.4′W. Olive black (5 Y 2/1) and grayish red (5 R 3/2), peloidal and banded iron-formation. Dip 60° to the northeast.
- Sites 05 and 06. Ridge top 1.5 km west of Peter Lake on southwest-dipping limb of anticline, at

54°51.6′N, 66°46.8′W. Moderate reddish brown (10 R 4/6) to grayish red (5 R 3/2), jaspilitic, peloidal and banded iron-formation. Dip 60° to the southwest.

- Site 07. Ridge top 1.5 km west of Peter Lake on southwest-dipping limb of anticline, at 54°51.5′N, 66°46.6′W. Moderate reddish brown (10 R 4/6) to grayish red (5 R 3/2), massive and banded iron-formation. Dip 70° to the southwest.
- Sites 08 and 09. By road 1.6 km west of Attikamagen Lake, Iron Arm, at 54°53.4′N, 66°37.0′W. Reddish brown (10 R 4/4), dark gray (N3) and grayish black (N2), jaspilitic and cherty peloidal iron-formation. Dip 80° to the southwest.
- Site 10. Near road 1 km east of Squaw Lake, at 54°51.4′N, 66°48.5′W. Dark gray (N3) and moderate yellowish brown (10 YR 5/4), peloidal and banded iron-formation. Dip 45° to the northeast.
- Sites 11 and 12. Near road 1.2 km east of Squaw Lake, at 54°51.5′N, 66°48.7′W. Blackish red (5 R 3/2), locally jaspilitic, peloidal and banded iron-formation. Dip 60° to the northeast.
- Site 13. Flank of ridge 0.8 km west of Peter Lake on northeast-dipping limb of the same anticline as sites 05–07, at 54°51.6′N, 66°46.1′W. Reddish brown (10 R 4/4) and medium dark gray (N4), jaspilitic, peloidal and banded iron-formation. Dip 30° to the northeast.

4.2. Composition of Sokoman iron-formation

Dimroth and Chauvel (1973) described the petrography of the Sokoman Formation from the central New Québec Orogen. They identified a variety of textures, including micrite of iron silicate, iron carbonate and matrix chert; chert and carbonate cements; and aggregates of pellets, intraclasts, ooliths and pisoliths. The rocks contain iron oxide in four principal forms: (1) hematite dust (<1 μm) that provides the red pigment in the jaspilite; (2) microcrystalline hematite or specularite (~30 μm to 1 mm); (3) magnetite octahedra (~3 μm); (4) skeletal crystals of magnetite and/or hematite that are interstitial to quartz. The iron-formation has undergone extensive epigenetic alteration, with compaction and cementation during early diagenesis and recrystallization during late diagenesis and metamorphism. Dimroth and Chauvel (1973) recognized the hematite dust as the oldest iron oxide, and much of the specularite and microscopic magnetite in iron-rich 'metallic' bands as early diagenetic. Further crystallization of iron oxides occurred during regional metamorphism. Carbonates include microgranular siderite, which Dimroth and Chauvel (1973) thought may be primary, and early diagenetic, concretionary siderite, ankerite and ferriferous calcite. Other constituent minerals include late diagenetic minnesotaite and stilpnomelane, and riebeckite which they ascribed to regional metamorphism.

X-ray diffraction analysis of samples from our 13 sites showed that hematite is the dominant iron oxide mineral, and is present with magnetite in virtually all samples. Goethite occurs in various amounts in samples from sites 01 and 02, and siderite is an important constituent of samples from site 09. Other minerals identified in minor to trace amounts include stilpnomelane, chlorite, calcite, dolomite, ankerite and greenalite (?).

Major element analyses for samples of iron-formation from the 13 sites are given in Table 2. On average, the samples contain 51.76% Fe_2O_3T or 34.89% Fe (range 23.76–80.24% Fe_2O_3T or 16.01–54.08% Fe). All the samples contain >15% Fe of primary origin (depositional or diagenetic) and hence are iron-formation as defined by James (1966). The samples with the highest Fe content are ore grade, implying that the iron-formation at some of our sites has undergone substantial Fe-enrichment.

5. Rock magnetism

5.1. Laboratory techniques

Bulk susceptibility (k) was measured with a transformer bridge calibrated using paramagnetic salts (Ridley and Brown, 1980). The bridge has a sensitivity of 10^{-7} SI. Anisotropy of magnetic susceptibility (AMS) was measured for standard-sized paleomagnetic specimens (25 mm diameter × 20 mm high) using a Digico anisotropy delineator that is interfaced to a personal computer for data manipulation. The operating system automatically corrects results for the Digico calibration error noted by Veitch et al. (1983) and Hrouda et al. (1983).

Thermomagnetic curves and susceptibility versus temperature measurements were obtained using a

Table 2
Major element analyses for iron-formation from 13 sites in the Sokoman Formation, Schefferville–Knob Lake area, New Québec Orogen

	SK01a	SK02b	SK03a	SK04a	SK05b	SK06b	SK07c	SK08c	SK09a	SK09b	SK10a	SK11c	SK12b	SK13b
SiO_2	52.16	50.69	55.63	56.50	42.29	19.57	30.97	52.12	25.93	25.86	54.78	50.15	34.35	15.54
Al_2O_3	0.05	0.04	2.89	0.44	0.42	0.60	2.93	0.49	1.99	1.46	9.38	3.77	0.61	0.36
Fe_2O_3 (T)	44.24	48.78	33.17	40.76	46.26	75.36	62.40	44.91	65.34	54.92	23.76	41.01	63.51	80.24
MnO	0.04	0.03	0.36	0.07	0.55	0.67	0.21	0.14	0.45	2.25	0.19	0.09	0.06	0.35
MgO	0.10	0.06	4.09	0.93	0.73	0.88	0.64	0.50	2.01	2.10	2.05	1.16	0.25	0.28
CaO	0.00	0.00	0.10	0.12	4.75	0.86	0.03	0.42	0.14	0.54	0.14	0.04	0.02	0.99
Na_2O	0.00	0.00	0.00	0.00	0.02	0.00	0.10	0.06	0.01	0.01	1.04	0.31	0.00	0.03
K_2O	0.00	0.00	0.59	0.15	0.16	0.28	1.94	0.07	0.78	0.46	6.38	2.45	0.07	0.20
TiO_2	0.02	0.01	0.35	0.05	0.04	0.08	0.11	0.03	0.26	0.18	1.31	0.37	0.05	0.01
P_2O_5	0.03	0.01	0.06	0.02	0.03	0.05	0.03	0.02	0.03	0.05	0.11	0.04	0.04	0.03
SO_3	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.01	0.03	0.01	0.01	0.01	0.01
LOI	3.06	0.00	2.26	0.56	4.29	1.35	0.41	0.66	1.63	11.06	0.22	0.10	0.36	1.25
Total (%)	99.70	99.62	99.50	99.60	99.54	99.70	99.77	99.43	98.58	98.92	99.37	99.49	99.33	99.29

Sample numbers refer to site/sample. Analysis by X-ray fluorescence, Geology and Geophysics, University of Adelaide.

Petersen variable-field translation balance. This translation balance operates as a harmonic oscillator with damping in the forced oscillation mode. Fast Fourier transform of the signal enhances its sensitivity by orders of magnitude over conventional magnetic balances. The variable-field translation balance can easily detect diamagnetic materials. A furnace/cooling unit allows samples to be measured between −175 and 700 °C. Samples usually are small cylindrical plugs (5 mm diameter × 5 mm thick) although crushed samples (∼200 mg) also can be used if packed tightly enough. The small plugs were cored from off-cuts of standard-sized paleomagnetic specimens.

5.2. Results

AMS data for 245 specimens of the Sokoman Formation indicate a well preserved bedding-parallel fabric, suggesting minimal alteration of the magnetic mineralogy since deposition and/or a mimetic secondary magnetic mineralogy. The iron-formation has not been internally deformed since the magnetic mineralogy was established. Stereographic projections of the axes of site (unit specimen) mean maximum and minimum susceptibilities, before and after unfolding, show a marked reduction in scatter and an overall mean minimum axis not significantly different from the pole to bedding (Fig. 4). Bulk susceptibilities average 0.207 SI (ranging from 4.2×10^{-3} SI for site 01 to 0.760 SI for site 03), indicating an average magnetite content of 6% (ranging from of 0.1 to 22%), using the relationship of Puranen (1989). Puranen (1989) has shown empirically for many lithologies that $k = 3.47f$ SI, where f is the volume fraction of magnetite. After unfolding, the minimum axes are close to vertical whereas the maxima show a weak lineation to the northwest–southeast. There is some switching between the maximum (NW–SE) and intermediate (NE–SW) axes. The mean site anisotropy ratio (major/minor axial susceptibilities) of 1.110 ± 0.059 for these rocks is intermediate between ratios for Paleoproterozoic iron-formations such as the Frere Formation from the Earaheedy Basin, Western Australia (Williams et al., 2003), the Gunflint Formation, Ontario (Schmidt and Williams, 2003), and banded iron-formation (BIF) from the Hamersley Basin, Western Australia (Schmidt and Clark, 1994).

High-field magnetization versus temperature measurements using the variable-field translation balance is shown in Fig. 5 (a field or 'induction' of 800 mT was applied). For sample SK01b the presence of both magnetite and hematite is apparent from the Curie temperatures of ∼580 and ∼680 °C (Fig. 5a). The magnetite has been partially destroyed on cooling, as shown by the irreversible curve. For other samples magnetite was produced by heating, which sometimes led to demagnetization problems; similar results were reported for samples of the Gunflint Formation (see Schmidt and Williams, 2003). Although the magnetite signal dominates, hematite is volumetrically more important. This is also the case for sample SK02a, although here the hematite signal is even more obscured

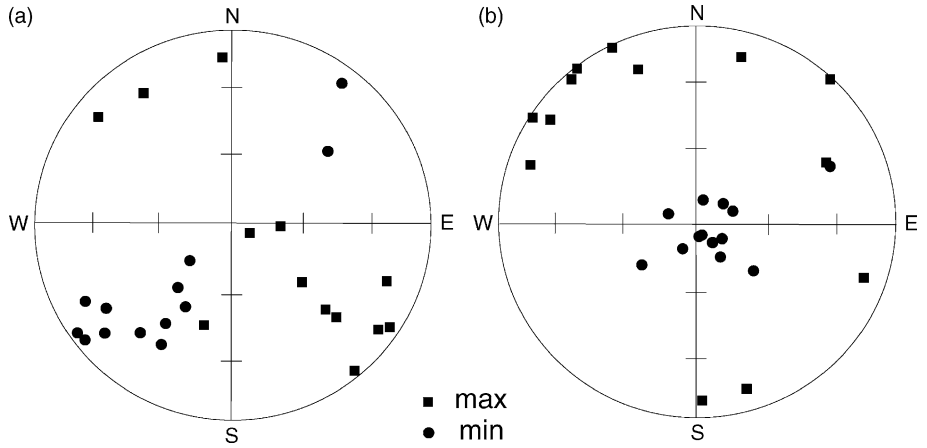


Fig. 4. Stereographic projections of AMS axes: (a) before and (b) after unfolding. The squares show axes of maximum susceptibility and the circles show axes of minimum susceptibility (all axes plotted on lower hemisphere).

by the much stronger signal from magnetite (Fig. 5d). The hysteresis loops shown in Fig. 5b and e have been corrected for paramagnetic contributions, which were determined by fitting linear trends to the data above 600 mT. Coercivities of remanence (H_{cr}) were determined from stepwise backfield demagnetization of saturation isothermal remanent magnetizations (IRMs), as shown in Fig. 5c and f. The parameters J_{rs}, J_s and H_c were taken from the corrected hysteresis loops. The J_{rs}/J_s ratios are 0.22 and 0.08 and the H_{cr}/H_c ratios are 2.5–3.0 for SK01b and SK02a, respectively. These values suggest reasonably stable magnetic carriers, although categorizing them as pseudo-single domain rather than multidomain is somewhat compromised by the presence both of magnetite and hematite. Nevertheless, the inferred stability of the remanence carriers is reflected in the paleomagnetic results given below.

6. Paleomagnetism

6.1. Laboratory techniques

Routine paleomagnetic laboratory methods (Collinson, 1983; Butler, 1992) were employed. Remanent magnetizations were measured using a 2G 755R three-axis cryogenic magnetometer interfaced to a computer. The demagnetizers used were a 2G 600 series AF demagnetizer and the CSIRO automated three-stage carousel furnace. The furnace is housed in a 4 m ten-coil Helmholtz set with automatic feedback maintaining zero-field (<5 nT). Following low-temperature pre-treatment (Schmidt, 1993), all specimens were subjected to stepwise thermal demagnetization. In addition, some specimens were subjected to stepwise AF demagnetization and AF pre-treatment prior to measuring after high-temperature thermal demagnetization steps.

Components of magnetization were isolated using an interactive version of Linefind (Kent et al., 1983), whereby linear segments are fitted to data points weighted according to the inverse of their measured variances. Excluding premature directional instability evidently caused by the oxidation of siderite to magnetite in one sample, the highest temperature to which any particular component remains stable, i.e. the highest unblocking temperature observed, is generally taken as an indication of the Curie temperature of the mineral carrying that component.

6.2. Results

NRM intensities varied for 254 mA m^{-1} (254 μG) to 15,600 mA m^{-1} (0.0156 G) and averaged 3700 ± 400 (s.e.) mA m^{-1} (3700 ± 400 μG). Combined with bulk susceptibilities, the NRM values convert to Koenigsberger ratios ranging from 0.108 to 2.20 with a mean of 0.477 ± 0.042 (s.e.), assuming a nominal field of 40 A m^{-1} (0.5 Oe). NRM directions were

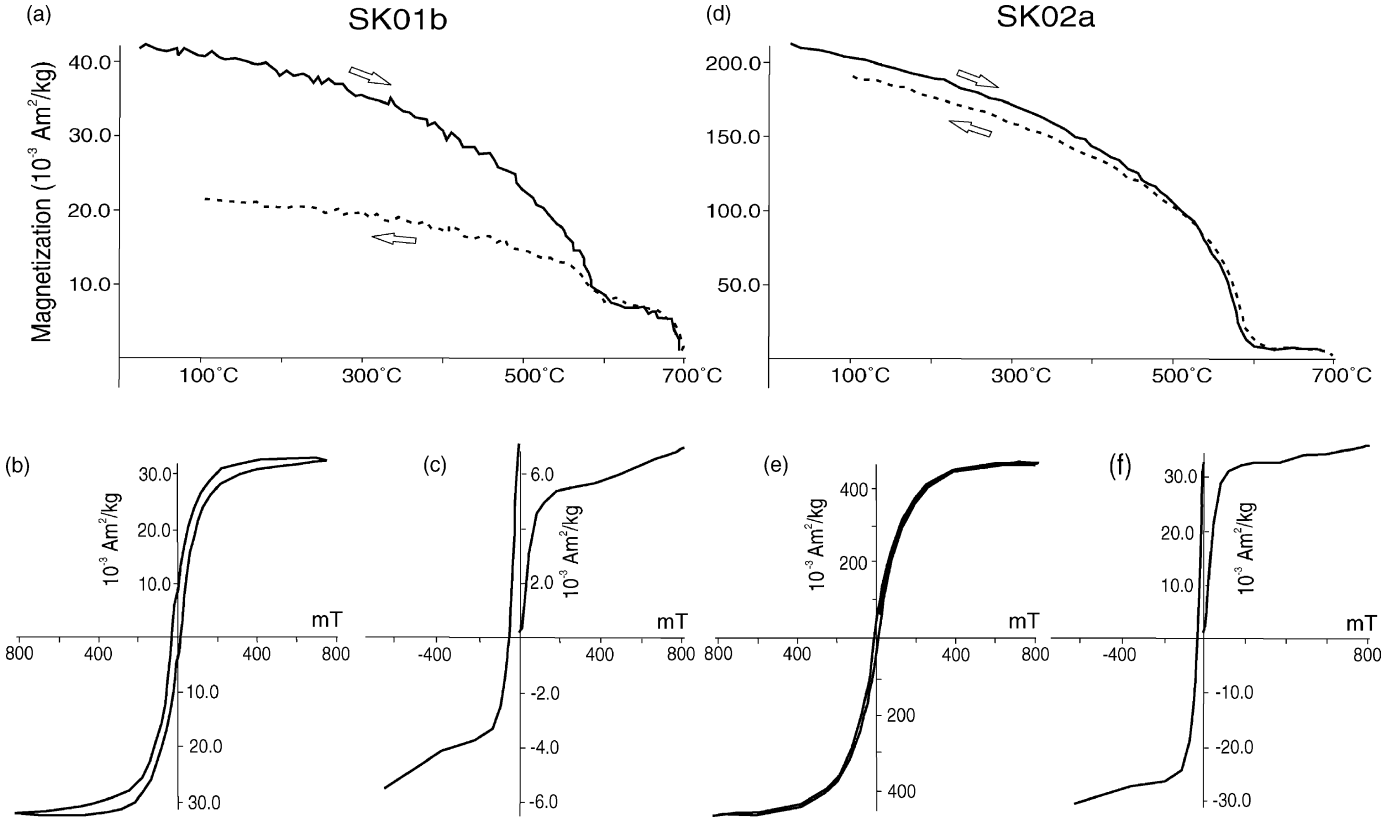


Fig. 5. Thermomagnetic curves and room-temperature hysteresis loops and IRM acquisition/decay curves for (a–c) SK01b and (d–f) SK02a. The magnetic field (induction) is given in mT and (specific) magnetization is in units $\times 10^{-3}$ A m^2 kg^{-1}. The hysteresis loops have been corrected for paramagnetic contributions by fitting lines to the data above 600 mT. Specimen numbers refer to site/sample/specimen.

streaked from shallow northeasterly or southwesterly toward the present field, which is steep down to the north–northwest (present field declination = 335°, inclination = 75°).

Typical orthogonal demagnetization plots are shown in Fig. 6 and paleomagnetic data determined herein are summarized in Table 3. Also shown in Table 3 is the percentage of reverse polarity magnetizations. The presence of reversals implies that the magnetizations were acquired over a long period of time, thus largely averaging secular variation. The scatter in the distributions probably results from some specimens possessing directions that are partly transitional. Such scatter is to be expected if the magnetizations are chemical remanent magnetizations acquired over long intervals.

Many specimens (e.g. Fig. 6a–c) revealed a soft component removed up to 400 °C that is directed southwesterly with a shallow inclination. Several specimens from site 09 acquired spurious components of magnetization with cooling after heating to 400–500 °C. Such unstable magnetization is ascribable to the presence of siderite (confirmed by X-ray diffraction analysis), which oxidizes to magnetite on heating (Pan et al., 2000).

The low-temperature component was followed by the demagnetization of a component up to the magnetite Curie temperature (580 °C). In some specimens (Fig. 6a–c) this component was directed westerly and steep down. Evidently this component is pre-folding in age and becomes shallow on application of full unfolding. In other specimens the magnetite component appears to be post-folding. The most stable component appears to be carried by hematite, which finally unblocked at ∼680 °C, and is directed to the north with a moderate down inclination. Fig. 6d shows that this component has dual polarity, although the two polarities were not prevalent enough to allow the execution of a reversal test. Other specimens (e.g. Fig. 6e and f) show a more or less single component from low-temperature to 680 °C, with only minor inflections at the temperatures that witnessed large angular disjunctions in the other examples. The magnetizations at these sites apparently are all post-folding in age.

From the above, therefore, some components apparently are pre-folding in age whereas other are syn- and post-folding. Figs. 7–9 show stereographic projections of specimen directions (a) before and (b) after unfolding for the low-temperature, magnetite and hematite components, respectively. Figs 8c and 9c show the directions for the magnetite and hematite components after varying amounts of unfolding to produce optimal convergence. Although the varying amounts of unfolding are somewhat arbitrary, they provide the simplest explanation for the observation that some site mean directions converge, whereas others diverge, to and from approximately the same magnetization axis. Full unfolding was applied to directions for sites 01–06 and 09, and for site 13 but only for the magnetite component. Partial unfolding was required for site 08, and for site 13 but only for the hematite component. No unfolding was applied for sites 07 and 10–12. Hence, the magnetization appears to be (1) pre-folding for sites 01–06 and 09, and for site 13 for magnetite; (2) syn-folding for site 08, and for site 13 for hematite; and (3) post-folding for sites 07 and 10–12. Furthermore, the results suggest that for site 13 the magnetite pre-dates the hematite.

The paleomagnetic components before and after applying the optimum amounts of unfolding discussed above are given in Table 3, which also shows results of applying fold tests (McFadden, 1990). The low-temperature component, which demonstrably fails the fold test, gives an in situ mean $D = 27.1°$, $I = 20.1°$ ($\alpha_{95} = 10.9°$) and a pole position (Pole C) of 40.6°S, 257.0°E (dp = 5.9°, dm = 11.4°). The magnetite component, after variable unfolding, gives a mean $D = 35.8°$, $I = 3.9°$ ($\alpha_{95} = 9.1°$) and a pole position (Pole B) of 29.6°S, 250.9°E (dp = 4.5°, dm = 9.1°). The hematite component, after variable unfolding, gives a mean $D = 40.0°$, $I = 1.6°$ ($\alpha_{95} = 11.5°$) and a pole position (Pole A) of 26.8°S, 247.0°E (dp = 9.3°, dm = 18.6°).

7. Discussion

7.1. Assessment of paleomagnetic data

Seguin (1976a) claimed to have identified a pre-folding NRM direction for Sokoman iron-formation with a paleopole position of 33°S, 256°E, similar to the poles found herein, and a post-folding NRM direction for combined data from diabase and iron-formation with a paleopole position of 16°N, 258°E. However, in his Table 2 the statistics of the combined AF and thermal results from the Sokoman Formation, after

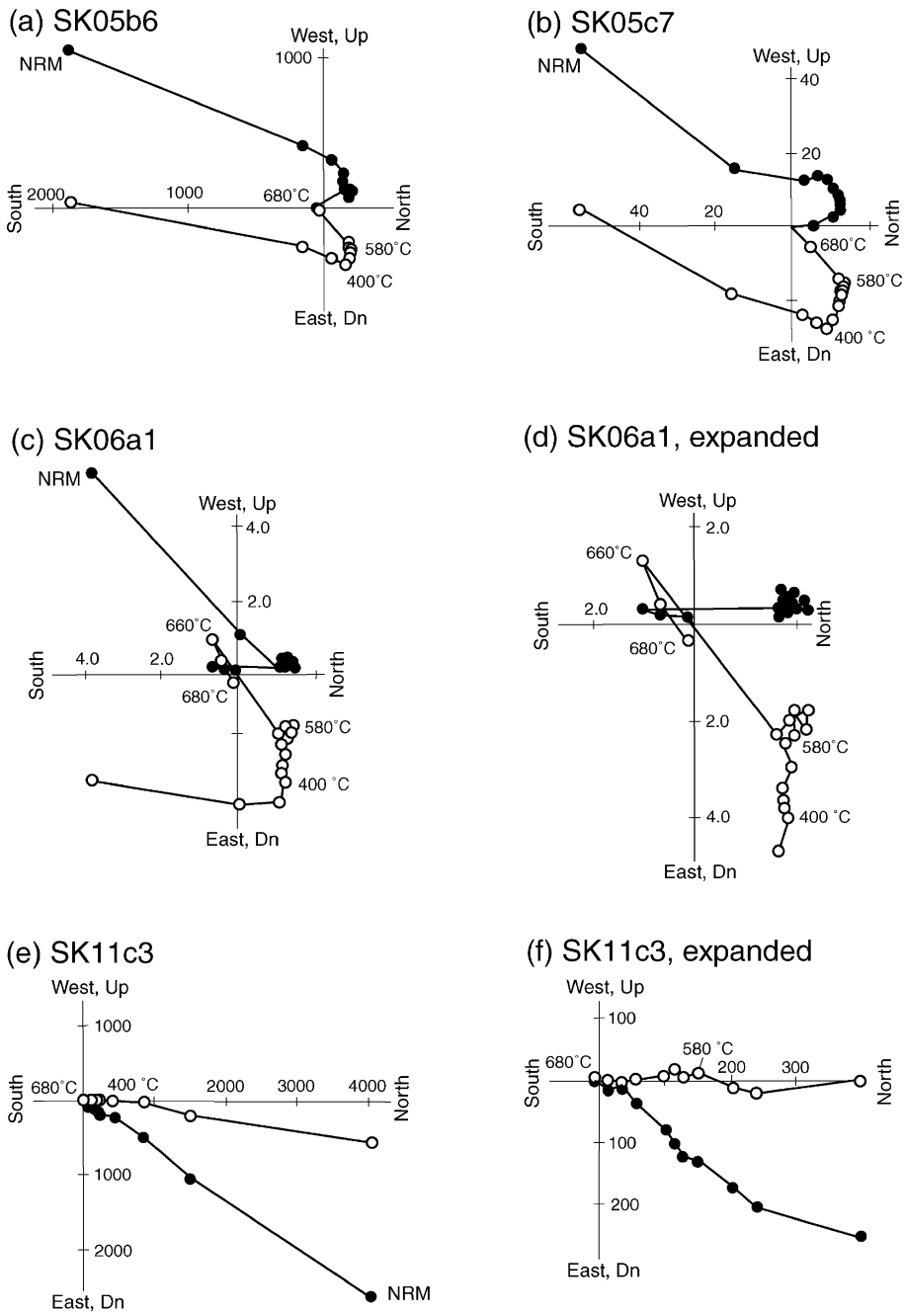


Fig. 6. Orthogonal projections of remanence vectors after stepwise thermal demagnetization. Solid circles plot on the horizontal plane and the open circles on the vertical plane. Units are $mA\,m^{-1}$ (μG). Specimen numbers refer to site/sample/specimen.

Table 3
Summary of paleomagnetic data determined herein for iron-formation from the Sokoman Formation, Schefferville–Knob Lake area, New Québec Orogen

Site	N	R%	D_h (°)	I_h (°)	K	α_{95} (°)	UF%	D_p (°)	I_p (°)	K	α_{95} (°)
Component C: maximum $T_{ub} \approx 400$ °C											
01	3	0	26.9	39.2	31.7	22.3	100	29.8	7.8	31.7	22.3
02	2	50	20.3	30.5	–	–	100	24.7	−0.8	–	–
05	11	100	205.3	−18.6	29.1	8.6	100	178.4	−77.8	29.1	8.6
07	9	100	207.7	−11.1	44.7	7.8	100	221.7	−80.7	44.7	7.8
10	15	0	35.4	5.6	15.8	9.9	100	32.7	−38.7	15.8	9.9
11	19	0	42.6	17.4	52.7	4.7	100	41.9	−42.5	52.7	4.7
13	7	0	20.8	17.0	14.5	16.4	100	22.0	−10.7	14.5	16.4
Mean	7	36	27.1	20.1	31.4	10.9		28.6	7.1	2.77	44.7[a]
Pole C (in situ): latitude = 40.6°S, longitude = 257.0°E (dp = 5.9°, dm = 11.4°)											
Component B: maximum $T_{ub} \approx 580$ °C											
01	6	0	20.8	38.8	33.0	11.8	100	25.0	8.1	33.0	11.8
02	7	71	199.6	−18.8	14.6	16.3	100	200.7	11.8	14.6	16.3
05	7	100	225.4	48.6	21.3	13.4	100	216.7	−8.2	21.3	13.4
06	4	100	194.5	46.7	90.8	9.7	100	198.3	−12.5	90.8	9.7
08	3	100	236.5	36.5	30.3	22.8	42	235.2	−3.3	30.3	22.8
09	6	50	0.9	−59.8	6.49	28.5	100	27.1	11.9	6.49	28.5
10	15	0	39.6	−5.6	12.3	11.4	0	39.6	−5.6	12.3	11.4
11	17	0	41.2	4.9	15.7	9.3	0	41.2	4.9	15.7	9.3
12	7	57	224.2	−11.5	5.73	27.6	0	224.2	−11.5	5.73	27.6
13	7	57	228.2	−29.7	23.7	12.7	100	229.7	4.7	23.7	12.7
Mean	10	44	34.9	−9.4	4.56	25.4		35.8	3.9	29.2	9.1[b]
Pole B (100% unfolding): latitude = 29.6°S, longitude = 250.9°E (dp = 4.5°, dm = 9.1°)											
Component A: maximum $T_{ub} \approx 680$ °C											
01	13	0	12.6	47.9	21.2	9.2	100	21.0	18.3	21.2	9.2
02	8	38	5.9	25.1	7.74	21.3	100	10.9	−1.5	7.74	21.3
07	2	100	225.4	8.5	–	–	0	225.4	8.5	–	–
08	2	50	59.5	−50.5	–	–	42	56.1	−10.8	–	–
11	6	0	38.4	−5.8	26.7	13.2	0	38.4	−5.8	26.7	13.2
12	3	100	252.5	0.3	6.39	53.4	0	252.5	0.3	6.39	53.4
13	6	67	214.5	−24.3	11.4	20.7	14	215.4	−19.8	11.4	20.7
Mean	7	51	39.1	5.6	4.62	31.5		40.0	1.6	11.5	18.6[b]
Pole A (100% unfolding): latitude = 26.8°S, longitude = 247.0°E (dp = 9.3°, dm = 18.6°)											

T_{ub}, unblocking temperature; N, number of specimens or sites; R%, percentage of specimens or sites with reverse polarity; D_h, I_h, mean declination and mean inclination with respect to present horizontal; UF%, percentage of unfolding applied; D_p, I_p, mean declination and mean inclination with respect to the interpreted paleo-horizontal; K, precision parameter; α_{95}, semi-angle of cone of 95% confidence about mean direction; dp, dm, semi-minor and semi-major axes of polar error ellipse.

[a] Negative fold test at 99% confidence.

[b] Some sites are pre-folding, some syn-folding and some post-folding (see text).

unfolding, are internally inconsistent. For instance, for the combined unfolded result, $N = 50$ and $K = 30$ were given with $\alpha_{95} = 33°$. These values are contradictory. If the values for N and K are correct then α_{95} should be 3.7°. For the thermal results alone $N = 19$, $K = 8.7$ and $\alpha_{95} = 38°$. Interestingly, $N = 19$, $R = 8.7$ and $\alpha_{95} = 38°$ are self-consistent, suggesting that the 'K' column should in fact be the 'R' column. If this were the case then the distribution would be almost random (uniform), since the R_0 ($P = 0.05$) is just under 7 for $N = 19$ (Watson, 1956). However, this is the only data line that is better satisfied by

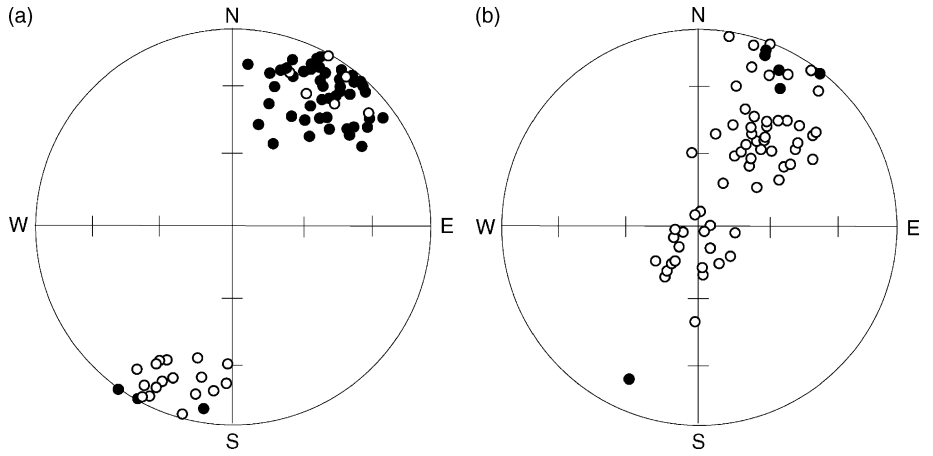


Fig. 7. Stereographic projections of components with low unblocking temperatures (<400 °C): (a) before unfolding, and (b) after unfolding. Solid circles plot on the lower hemisphere and open circles plot on the upper hemisphere.

swapping *R* for *K*. Virtually all the statistics in this table are erroneous, so the problem seems more extensive than typographical errors. Furthermore, because the pole position from the 'pre-folding' mean direction was apparently aberrant in terms of the pole path as defined then, Seguin (1976a) took a sub-set of the results and gave the poles quoted above. Details of the samples, or even of the new mean directions, are not given so the reader is left feeling somewhat apprehensive about the reliability of these two poles, and even of the whole study. We therefore reluctantly conclude that these results cannot be incorporated into a wider synthesis.

Seguin (1976b) gave a pole position for the Nimish Formation of 31°S, 239°E, after AF demagnetization of samples to 700 Oe and unfolding, that he related to the ~1750-Ma 'Hudsonian' orogeny. Although this pole is similar to pole positions found here for the Sokoman Formation, upon recalculating the mean direction quoted in his Table IV another disturbing result arises. If the sample directions listed in Table IV of Seguin (1976b) are correct, then a mean direction of $D = 211.7°$, $I = -17.9°$ ($\alpha_{95} = 38.5°$) follows, compared with his mean of $D = 226°$, $I = -18°$ ($\alpha_{95} = 13.7°$). Both the declination and the cone of confidence are in error, the latter perhaps being the more serious because it gives the false impression of a better defined pole position than in actuality. Indeed, if polarities are arbitrarily taken to minimize the scatter, which could be justified on the grounds that field reversals have been recorded, then a mean direction of $D = 217.3°$, $I = -14.3°$ ($\alpha_{95} = 24.0°$) is obtained, which still has quite a large error. The corresponding pole lies at 33.9°S, 246.8°E (dp = 12.5°, dm = 24.5°), which is in better agreement with the Sokoman poles herein than is the pole position that Seguin (1976b) calculated.

Turning to the pole position of 05°S, 268°E for the Nimish Formation that Seguin (1976b) gave after thermal demagnetization, yet another discrepancy is apparent. On inspection of his Table V, where the directions are listed and upon recalculation of the mean direction, it can be seen that his mean direction is seriously in error. In fact, the resultant vector for $N = 9$ is 1.318, indicating that the directions are random ($R_0 = 4.76$ for $N = 9$ and $P = 0.05$; Watson, 1956), that is, not significantly different from uniformly distributed. In this case it is meaningless to arbitrarily assign polarity in an attempt to reduce the scatter, on the grounds that reversals may be present, and any attempt to calculate a pole position is invalid.

The paleomagnetic pole position of 02°N, 205°E that Seguin (1976c) gave for the Menihek Formation and related gabbros also is difficult to reconcile with the mean directions given in his Table 2. By assuming the unit refers to the site, $N = 21$, which gives a mean direction of $D = 266.3°$, $I = 9.3°$ ($\alpha_{95} = 11.0°$) and a corresponding pole at 1.8°N, 208.6°E (dp = 5.6°, dm = 11.1°). In all of the above studies by Seguin there is a disconsolate lack of diagrams that

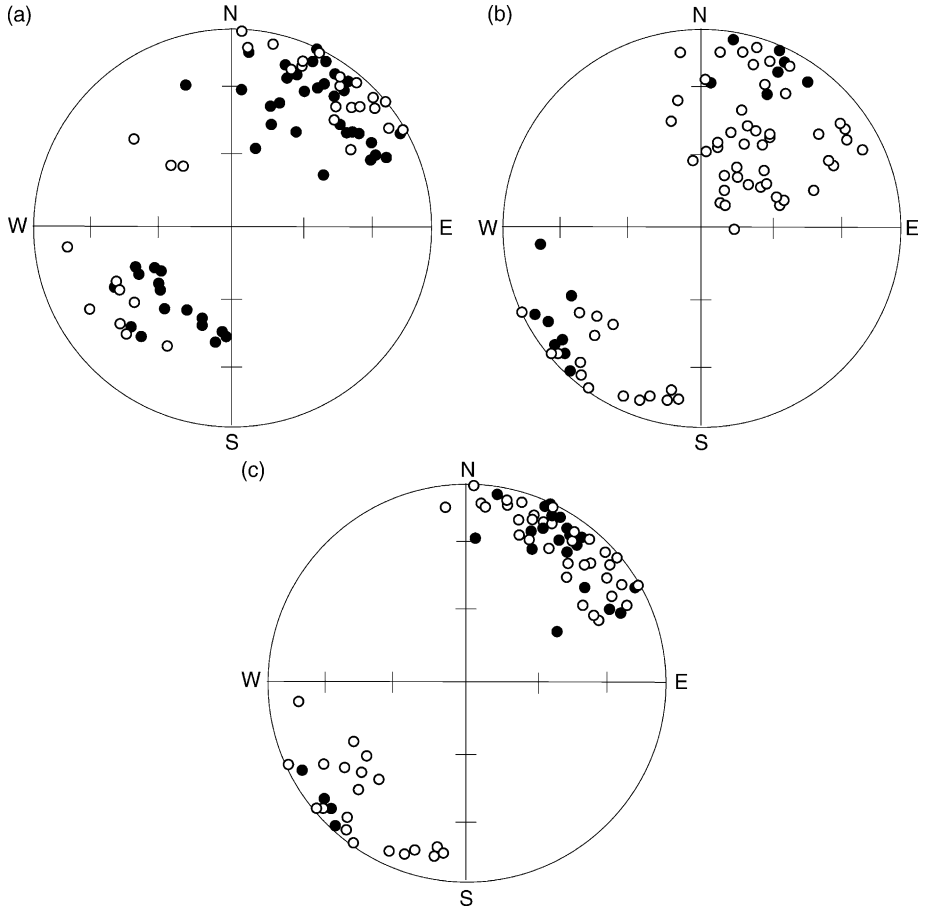


Fig. 8. Stereographic projections of components with magnetite unblocking temperatures (∼580 °C): (a) before unfolding, (b) after full unfolding, and (c) after variable unfolding to optimize convergence. Solid circles plot on the lower hemisphere and open circles plot on the upper hemisphere.

might overcome the many contradictions in the tabulated data. Unfortunately, it is not possible to view these studies as anything other than reconnaissance. Likewise, the results of Seguin (1979) cannot be interpreted because the reported statistics are inconsistent and the published diagrams are insufficient to allow an assessment of the data.

Schmidt (1980), in a reassessment of the reconnaissance study of Park (1977), argued on the basis of paleomagnetic studies of correlative successions from Cape Smith, the Richmond Gulf (Fujiwawa and Schwarz, 1975; Schwarz, 1976, 1977) and the Belcher Islands (Schmidt, 1980) that some magnetization ages may have been assigned to more ancient times than was justifiable. For instance, although Park (1977) tentatively suggested that the softest components of the Menihek Formation and Seward Subgroup of the Knob Lake Group may be ∼1.1 Ga, their similarity to the present field direction suggests that they might be recent viscous remanent magnetizations. In addition, his ∼1.9 Ga (?) directions for the Castigon carbonatitic complex in the central New Québec Orogen are very similar to overprint directions with negative fold tests from his other study areas. These overprints were thought to be 1.6–1.7 Ga (i.e. 'Hudsonian'). Indeed, notwithstanding the occurrence of dual polarity, intermediate stability components for the Menihek and Seward also are similar to these Hudsonian (?) overprints.

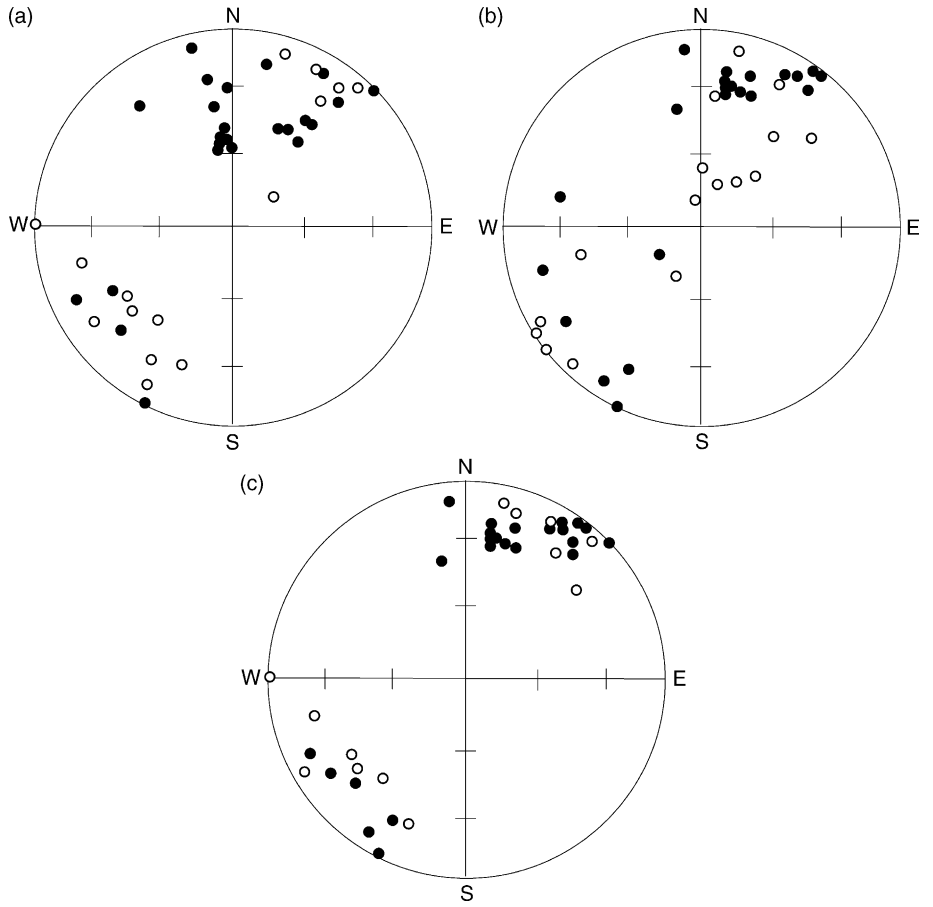


Fig. 9. Stereographic projections of components with hematite unblocking temperatures (∼680 °C): (a) before unfolding, (b) after full unfolding, and (c) after variable unfolding to optimize convergence. Solid circles plot on the lower hemisphere and open circles plot on the upper hemisphere.

Nevertheless, the most stable components of the Seward Subgroup (SA) bear some similarity to the Sokoman directions isolated by the present study. Although SA shows some streaking, which led Schmidt (1980) to suggest that it is composite, the 'end-member' directions are very close to the directions reported here in Table 3. Park (1977) also identified low inclination 'A' components in the Menihek Formation (MA) and the Retty peridotite (RA) of the overlying Montagnais Group. Combining all the A components for the Knob Lake and Montagnais groups given by Park (1977) yields a mean direction of $D = 187.7^\circ$, $I = 13.7^\circ$ ($\alpha_{95} = 20.2^\circ$) and a corresponding pole at 27.6°S, 284.3°E (dp = 10.5°, dm = 20.6°). Park (1977) sampled the Sokoman Formation at four sites and found scattered directions, which he ascribed to 'magnetic interactions in these concentrated oxide phases.'

In summary, previous paleomagnetic studies undertaken in the central New Québec Orogen suggest two episodes of overprinting. The older overprint, characterized by pre-, syn- and post-folding magnetization components, was hinted at in the work of Seguin (1976b) and again in that of Park (1977) and is the predominant component found in the present study. The younger overprint, although post-folding, does not appear to be related to deformation per se but probably is related to uplift and cooling.

The paleomagnetic poles determined herein are plotted in Fig. 10 where they are compared to the 'Hudsonian' pole from Seguin (1976b) and the mean

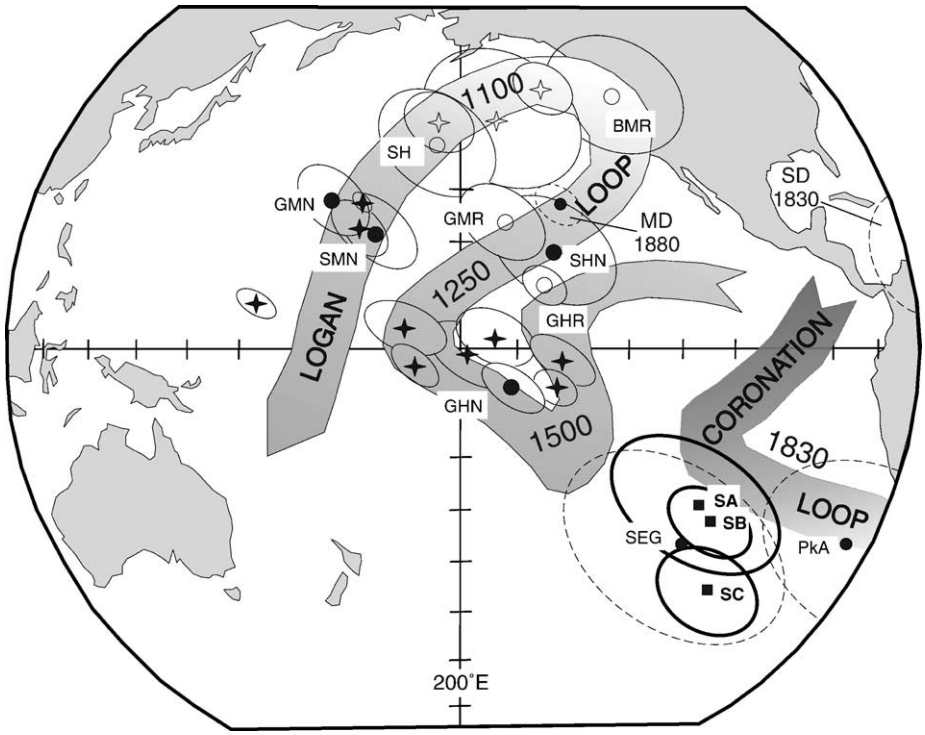


Fig. 10. Paleomagnetic poles for the Sokoman Formation determined herein, compared with Laurentian poles thought to be about the same age (1840–1829 Ma) and with the Coronation Loop for ~1850–1800 Ma (Irving and McGlynn, 1981). Poles SA, SB and SC refer to Sokoman A, B and C, and SEG and PkA are poles from the New Québec Orogen found by Seguin (1976b) and Park (1977), respectively, as discussed in the text. SD refers to the 1.83-Ga Sparrow dykes (Buchan et al., 2000) and MD to the 1.88-Ga Molson dykes (Halls and Heaman, 2000). The stars are key poles of Buchan et al. (2000). Also shown are the Logan Loop and new poles determined for Mesoproterozoic overprints from the 1.88-Ga Gunflint Formation (GMN, GHN, BMR, GMR, GHR) and poles for the ~1.5-Ga Sibley Group (SMN, SHN, SH) in Ontario (for details see Schmidt and Williams, 2003).

A pole from Park (1977), discussed above. Poles from the Gunflint Formation (Schmidt and Williams, 2003), which was overprinted during the interval ~1.5–1.0 Ga, and poles that form the Logan Loop are distinctly different from the Sokoman Formation poles. Such differences are consistent with the Sokoman Formation poles dating from the earlier time of 1.84–1.83 Ga, when deformation occurred in the New Québec Orogen.

Also shown in Fig. 10 is the pole for the 1.83-Ga Sparrow dykes (SD) (Buchan et al., 2000) that could be of a similar age to the Sokoman poles, although Buchan et al. (2000) stated that the quality of the SD pole is not comparable to that of key poles. Fig. 10 also includes the pole for the Molson dyke swarm (MD), which Halls and Heaman (2000) argued is a key pole dating from 1.88 Ga. (The proximity of the MD pole to the GMR and SHN poles does not imply they are coeval, because the latter two poles are for Mesoproterozoic overprints associated with strong negative field-tests; Schmidt and Williams, 2003.) Both the MD and the SD poles are distant from those found herein for the central New Québec Orogen. The Molson dyke swarm occurs along the western margin of the Superior Province, whereas the Sparrow dykes are from the Slave Province to the northwest. The paleogeography of North American cratons (structural provinces) is poorly constrained for the interval of interest here, but it seems unlikely that prior to ~1.8 Ga the cratons already were in their present relative positions. Therefore, it should not be surprising that poles of similar age from different

cratons or orogens are dispersed, and it may be suggested that the Sparrow, Molson and Sokoman directions were acquired before final convergence. An interval of rapid apparent polar wander between the MD pole and the Sokoman poles seems less likely. Interestingly, Rae-Superior convergence and collision at 1840–1829 Ma (Machado, 1990) was penecontemporaneous with the assembly by ∼1830 Ma of the West Australian, North Australian and South Australian cratons through collisions of Archean cratons (Myers et al., 1996). This interval may have witnessed widespread cratonic assembly that preceded Rodinia.

The 'Hudsonian' pole from Seguin (1976b) is seen to be very similar to our findings, whereas the mean A pole we have calculated from Park (1977) falls ∼20° to the east. Interestingly, the SA pole of Park (1977) plots at about 28°S, 267°E, closer to our results. However, the data of Seguin (1976b) and Park (1977) are not definitive and can only be viewed as being consistent with, rather than supporting, the directions found herein.

Fig. 10 also shows the older part of the Coronation Loop (Irving and McGlynn, 1981), which was thought to be ∼1850–1800 Ma, bracketing the age of deformation of 1840–1829 Ma in the New Québec Orogen (Machado, 1990). The applicability of the Coronation Loop to Laurentia as a whole has been questioned (e.g. Buchan et al., 1993) on the basis of key poles from dyke swarms. Nevertheless, the magnetizations determined for the Sokoman Formation are closely associated with deformation so it may be suggested that this part of the Coronation Loop is applicable to the New Québec Orogen.

7.2. Implications for the origin of the iron oxide orebodies

Our paleomagnetic results suggest that the episode of Fe-enrichment up to ore grade that affected iron-formation of the Sokoman Formation in the Schefferville–Knob Lake area occurred during late Paleoproterozoic deformation in the New Québec Orogen. This episode of Fe-enrichment is ascribable to the circulation of hydrothermal fluids related to orogenesis, and we suggest it contributed to the development of the hematitic orebodies in the region. Our data imply that late Paleoproterozoic deformation in the Schefferville–Knob Lake area occurred at a paleolatitude of ∼3–5°.

Models for enrichment proposed for the giant microplaty hematite orebodies hosted by early Paleoproterozoic (2.5 Ga) BIFs in the Hamersley Basin, Western Australia, appear relevant here. Morris (1980, 1985) postulated that following exposure of fresh BIF during initial uplift later in the Paleoproterozoic (∼2 Ga), iron was concentrated at depth by supergene processes. These processes involved the concentration of iron by leaching of silicates and the replacement of silicates by hydrous iron oxides, mostly goethite, with the mobile iron being derived from weathering of the BIF. In areas of favorable structure the processes penetrated deeply enough for the iron oxides to have avoided later erosion. The hydrated oxides were largely converted to hematite at temperatures of ∼100 °C during subsequent burial. Leaching of goethite following final uplift and exposure led to some secondary upgrading of old orebodies. An important point regarding paleomagnetic data for the Hamersley BIFs and iron-ores is that similar directions of magnetization are carried both by goethite-rich ore and BIFs (Schmidt and Clark, 1994), which implies that the magnetizations for the goethite-rich ore and BIFs are close in age. Although Morris (1980, 1985) concluded that a major interval of supergene enrichment took place during the Mesozoic–Cenozoic, the paleomagnetic evidence implies enrichment occurred early. Geologic mapping, basin analysis and calculated fluid compositions indicate that orebodies of microplaty hematite, and possibly martite–goethite, in the Hamersley Basin formed through the regional circulation of hydrothermal fluids, including heated meteoric water, during deformation at 2.4–2.2 Ga (Powell et al., 1999).

The age of the soft, goethitic orebodies in the Sokoman Formation remains open. The proposed Mesozoic age for these orebodies (Stubbins et al., 1961; Gross, 1968), as distinct from a likely Paleoproterozoic development of hematitic orebodies that is implied by our study, could be tested by paleomagnetism. The Schefferville–Knob Lake area was at paleolatitudes of ∼30°N in the Late Triassic and ∼40–50°N in the Jurassic and Cretaceous (see Smith et al., 1981; Scotese et al., 1988), in contrast to its equatorial paleolatitude during late Paleoproterozoic deformation. Samples from our sites 01 and 02 contain goethite but

give no indication of a Mesozoic paleomagnetic direction, which may cast doubt on a Mesozoic age for the goethitic orebodies. Paleomagnetic study of the goethitic orebodies in the New Québec Orogen may indicate whether they are indeed Mesozoic or date from an earlier episode of Fe-enrichment possibly related to orogenesis.

8. Conclusions

The magnetizations we have found in iron-formation from the 1.88-Ga Sokoman Formation in the Schefferville–Knob Lake area of the central New Québec Orogen, northern Québec, are pre-, syn- and post-folding in age. The softest component (C) with unblocking temperatures below 400 °C is entirely post-folding, whereas the components carried by magnetite (B) and hematite (A) are pre-, syn- and post-folding. The close association of the remanent magnetizations with deformation implies they are connected. The pole positions corresponding to these magnetizations are: Pole C, 40.6°S, 257.0°E (dp = 5.9°, dm = 11.4°); Pole B, 29.6°S, 250.9°E (dp = 4.5°, dm = 9.1°); and Pole A, 26.8°S, 247.0°E (dp = 9.3°, dm = 18.6°). We have calculated similar pole positions for rocks contiguous with the Sokoman Formation from tabulated directions given in earlier studies: data in Seguin (1976b) gives a pole position for the Nimish Formation of 33.9°S, 246.8°E (dp = 12.5°, dm = 24.5°), and combining all the A components for the Knob Lake and Montagnais groups given by Park (1977) yields a pole position of 27.6°S, 284.3°E (dp = 10.5°, dm = 20.6°).

Our paleomagnetic findings and the lack of Mesozoic remanence directions in the iron-formation, including goethitic rocks, suggest that the iron oxide orebodies in the Schefferville–Knob Lake area formed early. We conclude that an episode of Fe-enrichment resulting from the circulation of hydrothermal fluids related to orogenesis at 1.84–1.83 Ga contributed to the development of the hematitic orebodies in the region. This process is similar to the model for synorogenic enrichment proposed for the giant microplaty hematite orebodies hosted by early Paleoproterozoic banded iron-formation in the Hamersley Basin, Western Australia.

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